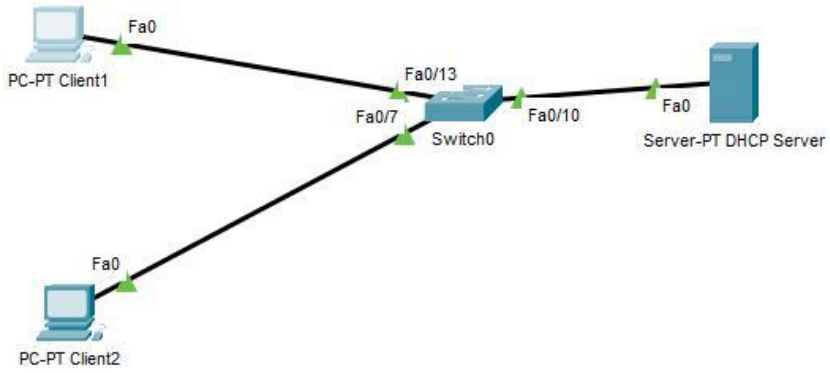
1) In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

1. Network topology



3. Configure the DHCP server

Click Server0 → Desktop → IP Configuration.

Enter the following static IP settings:

Setting	**Value**
IP Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

For this simple LAN, the gateway can be left blank because no router is connected.

Now configure DHCP:

1. Click Server0 → Services → DHCP.
2. Switch DHCP service to ON.
3. Enter these settings:

DHCP setting	**Value**
Pool Name	Office LAN
Default Gateway	192.168.1.1
DNS Server	8.8.8.8
Start IP Address	192.168.1.100

Subnet Mask	255.255.255.0
Maximum Users	50

4. Click Add or Save, depending on your Packet Tracer version.

Configurar servidor DHCP en Packet Tracer

byspel.com

4. Configure PC0 and PC1

For PC0:

1. Click PC0.
2. Open Desktop → IP Configuration.
3. Select DHCP.

For PC1, repeat the same steps.

Both PCs should automatically receive IP addresses from the DHCP server.

5. Verify IP addresses

Open Desktop → Command Prompt on each PC.

Type:

ipconfig

Expected result:

Device	IP address	Subnet mask
Server0	192.168.1.1	255.255.255.0
PC0	192.168.1.100	255.255.255.0
PC1	192.168.1.101	255.255.255.0

Expected result: DHCP working

6. Test connectivity

On PC0, open Command Prompt and enter:

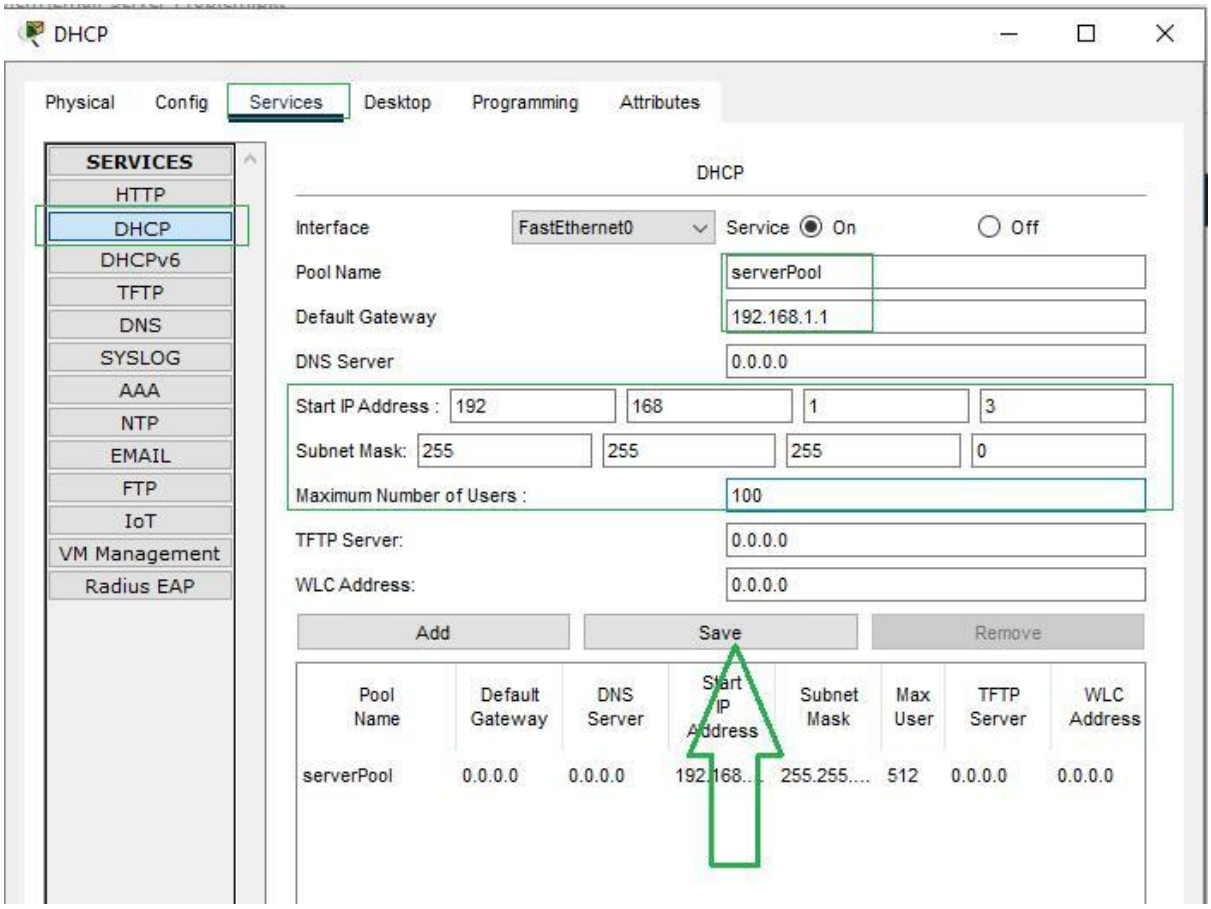
**2) Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.

Hint: Use arrows to show the direction of each message between client and server.</em**

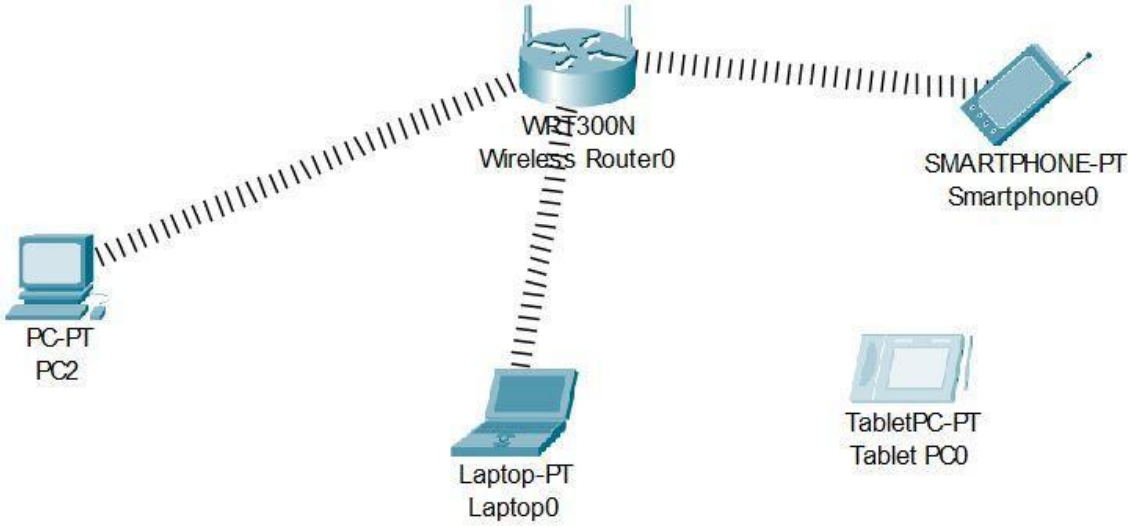
Explanation of four steps

Step	Sender → Receiver	Description
Discover	Client → Server	Client searches for a DHCP server.
Offer	Server → Client	Server offers an available IP address.
Request	Client → Server	Client requests the offered IP address.
Acknowledge	Server → Client	Server confirms the IP address assignment.

3) Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.



4) Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.



Required devices:

- **1 Wireless Router (WiFi Access Point)**
- **1 DHCP Server**
- **1 Laptop**
- **1 PC (representing a smartphone)**

2. Connect the devices

1. **Place the wireless router and DHCP server.**
2. **Connect the DHCP server to the wireless router using a Copper Straight-Through cable.**
3. **Connect the laptop and PC wirelessly to the router.**
4. **Configure the PC as a smartphone simulation.**

3. Configure the DHCP server

Click Server → Desktop → IP Configuration.

Configure the server with a static IP:

Setting	Value
IP Address	192.168.10.2
Subnet Mask	255.255.255.0

Default Gateway 192.168.10.1

Now open Services → DHCP and configure:

DHCP setting	Value
DHCP Service	ON
Pool Name	RestaurantWiFi
Default Gateway	192.168.10.1
DNS Server	8.8.8.8
Start IP Address	192.168.10.100
Subnet Mask	255.255.255.0
Maximum Users	50

Click Add/Save.

4. Configure wireless router

Open Wireless Router → GUI → Setup.

Configure:

Setting	Value
Router IP	192.168.10.1
Subnet Mask	255.255.255.0
DHCP Server	OFF (if external DHCP server is used)

Under Wireless settings:

- **Network Name (SSID): Zomato_Restaurant**
- **Security: WPA2-PSK**
- **Password: Zomato@12345**

Save the settings.

5. Connect laptop and smartphone

Laptop:

1. **Open Laptop → Desktop → PC Wireless.**
2. **Select Zomato_Restaurant.**
3. **Enter the WiFi password.**
4. **Open Desktop → IP Configuration.**

5. **Select DHCP.**

Smartphone (generic PC):

1. **Use a wireless-capable PC or install a wireless network adapter if supported.**
2. **Connect to Zomato_Restaurant.**
3. **Enter the WiFi password.**
4. **Select DHCP under IP Configuration.**